

Failure of GCH on a measurable with the Ultrapower Axiom

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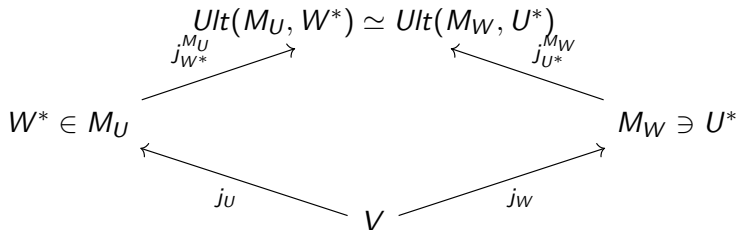
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The Ultrapower Axiom

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The Ultrapower Axiom (UA)

Let U, W be a pair of uniform, σ -complete ultrafilters. Then there exist uniform, σ -complete ultrafilters $U^* \in M_W, W^* \in M_U$ such that $M_{U^*}^{M_W} = M_{W^*}^{M_U}$ and $j_{U^*}^{M_U} \circ j_W = j_{W^*}^{M_W} \circ j_U$.



Consequences of UA

(Goldberg) The following are consequences of the Ultrapower Axiom:

- ① The Mitchell order is linear.
- ② Assume that κ is the least measurable cardinal, and let U be the unique normal measure on κ . Then every nontrivial, σ -complete ultrafilter on κ is isomorphic to U^n for some $n < \omega$.
- ③ The least Strongly Compact cardinal is Supercompact. Moreover, every Strongly Compact cardinal is either Supercompact, or a measurable limit of Supercompacts.
- ④ Assume that κ is Supercompact. Then $\text{GCH}_{\geq \kappa}$ holds.

Question (Goldberg)

Is UA consistent with the failure of GCH on a measurable cardinal κ ?

- Starting from an inner model with suitably large cardinals, one can achieve the failure of GCH on a measurable, by performing an iterated forcing of length $\kappa + 1$.
- However, many known methods to do so fail to preserve the UA, since they violate the linearity of the Mitchell order.
- Other methods preserve the linearity of the Mitchell order, but not the Ultrapower Axiom.

- **Friedman - Magidor, 2009:** Starting from a cardinal κ which is $(\kappa + 2)$ -strong, forced the failure of *GCH* at the measurable κ , with a unique normal measure on it.
- **Ben-Neria - Gitik, 2017:** Starting from a cardinal κ with $o(\kappa) = \kappa^{++}$, forced the failure of *GCH* at κ , with a unique normal measure on it.
- **Apter - Cummings, 2023:** Starting from a strong cardinal κ , forced the failure of *GCH* on a measurable cardinal, preserving the strongness of κ and the linearity of the Mitchell order on it. (since κ remains strong in the generic extension, it carries $(2^\kappa)^+$ -many normal measures).

The Main Result

Theorem (Ben-Neria, K.)

Let $L[\vec{E}]$ be the minimal inner model for the existence of a measurable cardinal κ carrying a (κ, κ^{++}) -extender. Then there exists a forcing extension in which GCH fails at κ and UA holds.

- The forcing is a variation of the Friedman-Magidor forcing.
- We present below two blueprints:
 - The first, **the Friedman-Magidor blueprint**, extracts the properties of the Friedman Magidor forcing, that ensure that GCH breaks at a measurable cardinal κ carrying a unique normal measure U .
 - The second, **the Kunen-like blueprint**, extends the Friedman-Magidor blueprint by adding two additional requirements, ensuring that every σ -complete ultrafilter is Rudin-Keisler equivalent to U^n , for some $n < \omega$.
- An important part of the work (omitted here) is the construction of a forcing notion that fulfills the Kunen-like blueprint. One of the main features of the construction is the integration of fine-structure into it.

The Friedman-Magidor blueprint

The Friedman-Magidor blueprint

Work in $L[\bar{E}]$. Assume that $\langle \mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha : \alpha \leq \kappa + 1 \rangle$ is an iterated forcing. We say that it fulfils the Friedman-Magidor blueprint if it satisfies the following properties:

The Friedman-Magidor blueprint

- ① For each $\alpha \leq \kappa$, the $\mathbb{P}_\alpha$ -name $\dot{\mathbb{Q}}_\alpha$ is the trivial poset unless α is Mahlo.
- ② For every Mahlo cardinal $\alpha \leq \kappa$, $0_{\mathbb{P}_\alpha}$ forces $\dot{\mathbb{Q}}_\alpha$ satisfies the following requirements:
 - $\dot{\mathbb{Q}}_\alpha \in V_\kappa$ (unless $\alpha = \kappa$), has size $|\dot{\mathbb{Q}}_\alpha| = \alpha^{++}$ and adds α^{++} new subsets to α ,
 - $\dot{\mathbb{Q}}_\alpha$ is α -distributive, satisfies the α^{++} .c.c, and preserves the Mahloness of α .
 - $\dot{\mathbb{Q}}_\alpha$ satisfies the α Continuous-Fusion property*,
 - $\dot{\mathbb{Q}}_\alpha$ is self coding**.

*Fusion arguments played an important role in the Friedman-Magidor theorem. The " α Continuous-Fusion property" and the "Iteration-Fusion property" (which will be mentioned below), are abstract properties allowing to perform those arguments.

**The notion of "Self-Coding forcing" will be properly defined below.

The Friedman-Magidor blueprint - continued

- ③ For every regular cardinal $\alpha \leq \kappa$,
 - $\mathbb{P}_\alpha$ has the α Iteration-Fusion property,
 - $0_{\mathbb{P}_\alpha} \Vdash (\mathbb{P}/\mathbb{P}_\alpha)$ is α -distributive.
- ④ The iteration $\mathbb{P} = \mathbb{P}_{\kappa+1}$ is uniformly definable in the sense that there is a parameter-free formula $\varphi(x)$ such that, for every $\alpha \leq \kappa$, $\varphi(\alpha)$ defines $\mathbb{P}_{\alpha+1}$ and some relevant parameters*** correctly over every inner Model that contains $H_{\alpha^{++}}$.

***Other than $\mathbb{P}_{\alpha+1}$ itself, the formula should also define some parameters related to Self-Coding, and the $\mathbb{P}_{\alpha+1}$ -names for the α^{++} -many subsets added to α by $\mathbb{Q}_\alpha$.

The Friedman-Magidor blueprint

Theorem (Friedman-Magidor)

Assume $L[\bar{E}]$ is the minimal inner model with a measurable cardinal κ carrying a (κ, κ^{++}) extender, and let $\mathbb{P} = \mathbb{P}_{\kappa+1}$ be an iteration satisfying the Friedman-Magidor blueprint. Then $\mathbb{P}$ forces that κ is the least measurable, $2^\kappa = \kappa^{++}$, and there exists a unique normal measure on κ .

Main ideas in the proof that κ carries a unique normal measure

- Assume that $G \subseteq \mathbb{P}_{\kappa+1}$ is $\mathbb{P}_{\kappa+1}$ -generic over $V = L[\bar{E}]$, and $U \in V[G]$ is a normal measure on κ .
- Let $j_U: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. It's enough to show the following:

Properties ensuring that U is uniquely determined:

- 1 $M = M_E$, $j_E \subseteq j_U$, where E is the top (κ, κ^{++}) -extender in $L[\bar{E}]$.
- 2 H is unique, in the sense that it is the only $j_E(\mathbb{P}_{\kappa+1})$ -generic set over M in $V[G]$, satisfying $j_E[G] \subseteq H$.

$$① \quad M = M_E, j_E \subseteq j_U$$

The proof that $j_U \restriction V = j_E$ and $M = M_E$ goes as follows:

- $j_U \restriction V$ is an iterated ultrapower by extenders. Letting E_0 being the first extender applied in the normal iteration, we first show that $E_0 = E$ is the top (κ, κ^{++}) -extender.
- Indeed, otherwise,

$$\kappa < j_{E_0}(\kappa) \leq \kappa^{++} \leq (2^\kappa)^{M[H]}$$

contradicting the fact that $j_{E_0}(\kappa)$ is strongly inaccessible in $M[H]$.

- The Fusion properties of $\mathbb{P}_\kappa, \mathbb{Q}_\kappa$ can be used to approximate functions in $(\text{Ord}^\kappa)^{V[G]}$ by ground model functions; this fact can be used to deduce that $M = M_E$ and $j_U \restriction V = j_E$.

2 H is unique

The next step is to prove the uniqueness of $H \in V[G]$ which is $j_E(\mathbb{P})$ -generic over M_E , with $j_E[G] \subseteq H$. Very roughly, this is done as follows:

- By standard arguments, $H \restriction \kappa = G \restriction \kappa$.
- The fact that each $\dot{\mathbb{Q}}_\alpha$ is self-coding implies that $H(\kappa) = G(\kappa)$.
- The fact that $\mathbb{P}_\kappa$ satisfies the κ Iteration-Fusion property implies that $H(\kappa, j_E(\kappa))$ is generated from $j_E[G] \setminus \kappa + 1 = \{j_E(p) \setminus \kappa + 1 : p \in G \restriction \kappa\}$.
- The fact that each $\mathbb{Q}_\alpha$ satisfies the α C-Fusion property implies that $H(j_E(\kappa))$ is the unique $j_E(\dot{\mathbb{Q}}_\kappa)$ -generic in $V[G]$ that contains $\tilde{j}[G(\kappa)]$ (here, $\tilde{j}: V[G \restriction \kappa] \rightarrow M_E[j_E[G \restriction \kappa]]$ denotes the natural lift of j_E).

Below, we will elaborate on the definition of Self Coding and the proof that it implies that $H(\kappa) = G(\kappa)$.

Definition - Self-Coding

Let $\mathbb{Q}$ be a poset, $\vec{q} = \langle q_\alpha : \alpha < |\mathbb{Q}| \rangle$ be an enumeration of $\mathbb{Q}$ and $\vec{S} = \langle S_\alpha : \alpha < |\mathbb{Q}| \rangle$ be a sequence of pairwise almost-disjoint stationary subsets of λ for some regular $\lambda \leq |\mathbb{Q}|$.

We say that $\mathbb{Q}$ **admits self coding** via $\vec{q}, \vec{S}$ if and only if, for every $\tau < |\mathbb{Q}|$,

$$0_{\mathbb{Q}} \Vdash (\check{S}_{2\tau} \text{ is not stationary} \iff q_\tau \in \dot{G}_{\mathbb{Q}}) \text{ and}$$

$$0_{\mathbb{Q}} \Vdash (\check{S}_{2\tau+1} \text{ is not stationary} \iff q_\tau \notin \dot{G}_{\mathbb{Q}}).$$

where $\dot{G}_{\mathbb{Q}}$ denotes the canonical $\mathbb{Q}$ -name for the generic set.

In the blueprints, we require that each iterate $\mathbb{Q}_\alpha$ ($\alpha \leq \kappa$) satisfies Self-Coding. We also require that the lists $\vec{q}, \vec{S}$ witnessing that, have uniformly definable $\mathbb{P}_\alpha$ -names over $H_{\alpha^{++}}$.

The uniqueness of $H(\kappa)$

Using the self coding, we can prove that $H(\kappa) = G(\kappa)$.

- Assume that the self-coding of $\mathbb{Q}_\kappa$ is witnessed by sequences $\langle q_\alpha : \alpha < \kappa^{++} \rangle, \langle S_\alpha : \alpha < \kappa^{++} \rangle$.
- The same sequences witness the self coding of $\mathbb{Q}_\kappa$ over $M_E[G_{<\kappa}]$.
- Therefore, for every $\tau < \kappa^{++}$,

$$q_\tau \in G(\kappa) \iff V[G] \models S_{2\tau+1} \text{ is stationary} \implies \\ M_E[H] \models S_{2\tau+1} \text{ is stationary} \iff q_\tau \in H(\kappa).$$

The Kunen-like blueprint

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The Kunen-Like blueprint

The Kunen-Like blueprint is obtained from the Friedman-Magidor blueprint by adding the following assumptions:

- For every Mahlo cardinal $\alpha \leq \kappa$, $\dot{\mathbb{Q}}_\alpha$ adds an increasing sequence in $<_{Sing}^\alpha$ of length α^{++} .
(for $f, g \in \alpha^\alpha$, we say that $f <_{Sing}^\alpha g$ if there exists a club $C \subseteq \alpha$ such that for every singular $i \in C$, $f(i) < g(i)$).
- The self coding of $\dot{\mathbb{Q}}_\alpha$ is being done by sequences $\vec{q}^\alpha$ and $\vec{S}^\alpha$, where $\vec{S}^\alpha = \langle S_\tau^\alpha \mid \tau < \alpha^{++} \rangle$ consists of almost disjoint stationary subsets of $\alpha^+ \cap \text{Cof}(\rho_\alpha)$, for some regular cardinal $\rho_\alpha < \alpha$.

The main Theorem - proof sketch

① Let $G \subseteq \mathbb{P}$ be generic over $V = L[\bar{E}]$, where $\mathbb{P}$ is a forcing notion which satisfies the Kunen-Like blueprint.

② In $V[G]$, κ is a measurable cardinal, carrying a unique normal measure U .

We argue that any nontrivial σ -complete ultrafilter $W \in V[G]$ is RK -equivalent to U^n , for some $n < \omega$.

③ Identifying n : Let $j_W: V[G] \rightarrow M[H]$ be the associated ultrapower embedding. Then $j_W \upharpoonright V$ is an iterated ultrapower of $L[\bar{E}]$ by its extenders. Since $\mathbb{P}$ is ω_1 -distributive, the iteration must have a finite length - denote it by n .

In order to show that $W \equiv_{RK} U^n$, it's enough to prove:

① $M = \text{Ult}(V, E^n)$, where E is the top (κ, κ^{++}) -extender of $L[\bar{E}]$.

② H is unique, and is actually equal to $j_{U^n}(G)$.

For simplicity, we concentrate on the case where $n = 2$.

① The proof that $M = M_{E^2}$

- Assume that $j_W \upharpoonright V$ is a two step iteration, given in its normal form. Arguing as in the Friedman-Magidor blueprint, the first step must be an ultrapower with the (κ, κ^{++}) -extender E .
- Denote by $F \in \text{Ult}(L[\bar{E}], E)$ the second extender which is being applied.

Claim

$$F = j_E(E).$$

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Proof Sketch

Denote $\lambda = \text{crit}(j_F)$. Then $\lambda = j_E(\kappa)$ and $F = j_E(E)$, since:

$$(\lambda^{++})^{M_E} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{\text{Ult}(V[G], U)} \leq \left(\text{rank} \left(<_{\text{Sing}}^\lambda \right) \right)^{M[H]} < j_F(\lambda)$$

- The leftmost inequality follows since $\lambda \leq j_E(\kappa)$ and thus $j_E(\mathbb{P})$ adds an increasing $<_{\text{Sing}}^\lambda$ sequence of length $(\lambda^{++})^{M_E}$.
- For the middle inequality:
 - Since U is the unique normal measure on κ in $V[G]$ and $\text{crit}(j_W) = \kappa$, $U = \{X \subseteq \kappa : \kappa \in j_W(X)\}$.
 - Therefore, there exists an embedding $k: \text{Ult}(V[G], U) \rightarrow M[H]$ such that $j_W = k \circ j_U$. It could be shown that $\text{crit}(k) \geq \lambda$.
 - Thus $(\mathcal{P}(\lambda))^{\text{Ult}(V[G], U)} \subseteq (\mathcal{P}(\lambda))^{M[H]}$, which implies the inequality.
- The rightmost inequality follows since $j_F(\lambda)$ is inaccessible in M , and remains such in $M[H]$.

2 The proof that H is uniquely determined

The set $H \subseteq j_{E^2}(\mathbb{P})$ is the unique generic over M_{E^2} with $j_{E^2}[G] \subseteq H$. This is shown in the following steps:

- $H \restriction j_E(\kappa) = j_U(G \restriction \kappa)$: this follows as in the Friedman-Magidor blueprint.
- $H(j_E(\kappa)) = j_U(G)(j_E(\kappa))$: this uses the additional assumptions regarding the Self-Coding procedure in the Kunen-like blueprint (we will elaborate on this point).
- $H \restriction (j_E(\kappa), j_{E^2}(\kappa)]$ is uniquely determined: this follows by Fusion arguments.

Claim

$$H(j_E(\kappa)) = j_U(G)(j_E(\kappa)).$$

Proof Sketch

For simplicity, we denote $\kappa_1 = j_E(\kappa)$.

- ① It suffices to argue that $j_U[G(\kappa)] \subseteq H(\kappa_1)$.
- ② Fix the parameters witnessing the Self-Coding of $\mathbb{Q}_\kappa$: an enumeration $\langle q_\alpha : \alpha < \kappa^{++} \rangle$ of $\mathbb{Q}_\kappa$, and a list $\langle S_\alpha : \alpha < \kappa^{++} \rangle$ of pairwise almost disjoint stationary subsets of $\kappa^+ \cap \text{cf}(< \kappa)$.
- ③ Assume that for some $\alpha < \kappa^{++}$, $q_\alpha \in G(\kappa)$, but $j_U(q_\alpha) \notin H(\kappa_1)$.
Then:
 - In $V[G]$, $S_{2\alpha}$ is nonstationary in κ^+ , and $S_{2\alpha+1}$ is stationary in κ^+ .
 - In $M[H]$, $j_U(S_{2\alpha})$ is stationary in κ_1^+ , and $j_U(S_{2\alpha+1})$ is nonstationary in κ_1^+ .

Proof Sketch - Continued

- ④ Therefore, in order to derive a contradiction, we should prove that for every $S \subseteq \kappa^+ \cap \text{Cof}(< \kappa)$,
If S is a stationary subset of κ^+ in $V[G]$, then $j_U(S)$ is a stationary subset of κ_1^+ in $M[H]$.
- ⑤ Given a club $C \subseteq \kappa_1^+$ in $M[H]$, $(j_U)^{-1}[C]$ is a $< \kappa$ -club in κ^+ in $V[G]$. It intersects S , since S concentrates on ordinals of cofinality strictly below κ . Applying j_U on the intersection point yields that $C \cap j_U(S) \neq \emptyset$.

Realizing the Kunen-Like blueprint

- In the Friedman-Magidor forcing, the stationary sets witnessing the self-coding of $\mathbb{Q}_\alpha$ (for $\alpha \leq \kappa$ Mahlo) are contained in $\alpha^+ \cap \text{Cof}(\alpha)$. This implies that $\mathbb{Q}_\alpha$ is α -closed, which is instrumental to the fact that $\mathbb{P}_\kappa$ has distributive quotients and satisfies the κ -Iteration Fusion property.
- The Kunen-like blueprint requires the use of stationary subsets of $\alpha^+ \cap \text{Cof}(< \alpha)$. The forcings $\mathbb{Q}_\alpha$ are no longer closed; they still remain α -distributive, but there is no¹ clear iteration scheme of distributive forcings which behaves "nicely" (namely, has distributive quotients, and satisfies the Iteration-Fusion property).
- To solve this, the Fine-Structure of $V = L[\bar{E}]$ is utilized. In particular, the same theory behind the construction of square sequences in L is applied to carefully craft the stationary sets used for coding, so that the entire iteration has the desired properties.

¹By results of Gitik, distributive forcings could be realized as Prikry-type forcings, which have an iteration scheme that behaves "nicely". However, realizing distributive forcings as Prikry-type forcings requires stronger large cardinals than those at our disposal.

Questions

- Could $\text{UA} + \kappa$ is measurable $+ 2^\kappa = \kappa^{++}$ be achieved from $\text{co}(\kappa) = \kappa^{++}$?
- Could $\text{UA} + \kappa$ is measurable $+ 2^\kappa = \kappa^{++}$ be forced with a more complex measure structure? (for instance, several normal measures on κ , rather than a unique one).
- Could UA be hold with $2^\kappa = \kappa^{+3}$ where κ is measurable?

Thank you for your attention!